MACHINE LEARNING PROJECT • ECE

Zero-Reference Low-Light Video Enhancement

Code Link (Github)

Test Dataset Link

Zero-DCE · Unsupervised Image-to-Image Translation

7	8	100	4
Datasets	Curve Iters	Epochs	Loss Terms

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Problem Statement & Motivation

Poor Visibility

Low-light images suffer from noise, low contrast, and colour distortion — making them unusable for human viewers and computer vision systems alike.

Supervised Limits

Existing deep-learning methods need paired low/normal-light image datasets, which are expensive and difficult to capture in real-world conditions.

Video Flickering

Frame-by-frame enhancement causes temporal inconsistency — lights flicker and brightness jumps between consecutive frames.

Our Approach

- **Zero-Reference Learning**

  No paired data needed. The model trains on low-light images alone, guided purely by physics-inspired loss functions.

- **DCE-Net Architecture**

  A lightweight 7-layer CNN predicts pixel-wise, image-specific tonal curves — achieving real-time inference speed.

- **Temporal Smoothing**

  Curve maps are exponentially blended across frames (α=0.15) to eliminate flickering in video output.

- **Custom Loss Design**

  Four complementary losses — illumination smoothness, spatial consistency, colour constancy, and masked exposure — jointly guide training.

Dataset

7-dataset fusion — all low-light, no paired references required

LOL

Low-Light dataset
485 low/normal pairs
Indoor & outdoor

LOL-v2

Improved split
Real + synthetic
689 samples

LIME

10 challenging
low-light images
Benchmark set

LSRW

Large-Scale
Real-World low-light
Multi-scene

MEF

Multi-Exposure
Fusion benchmark
17 sequences

DICM

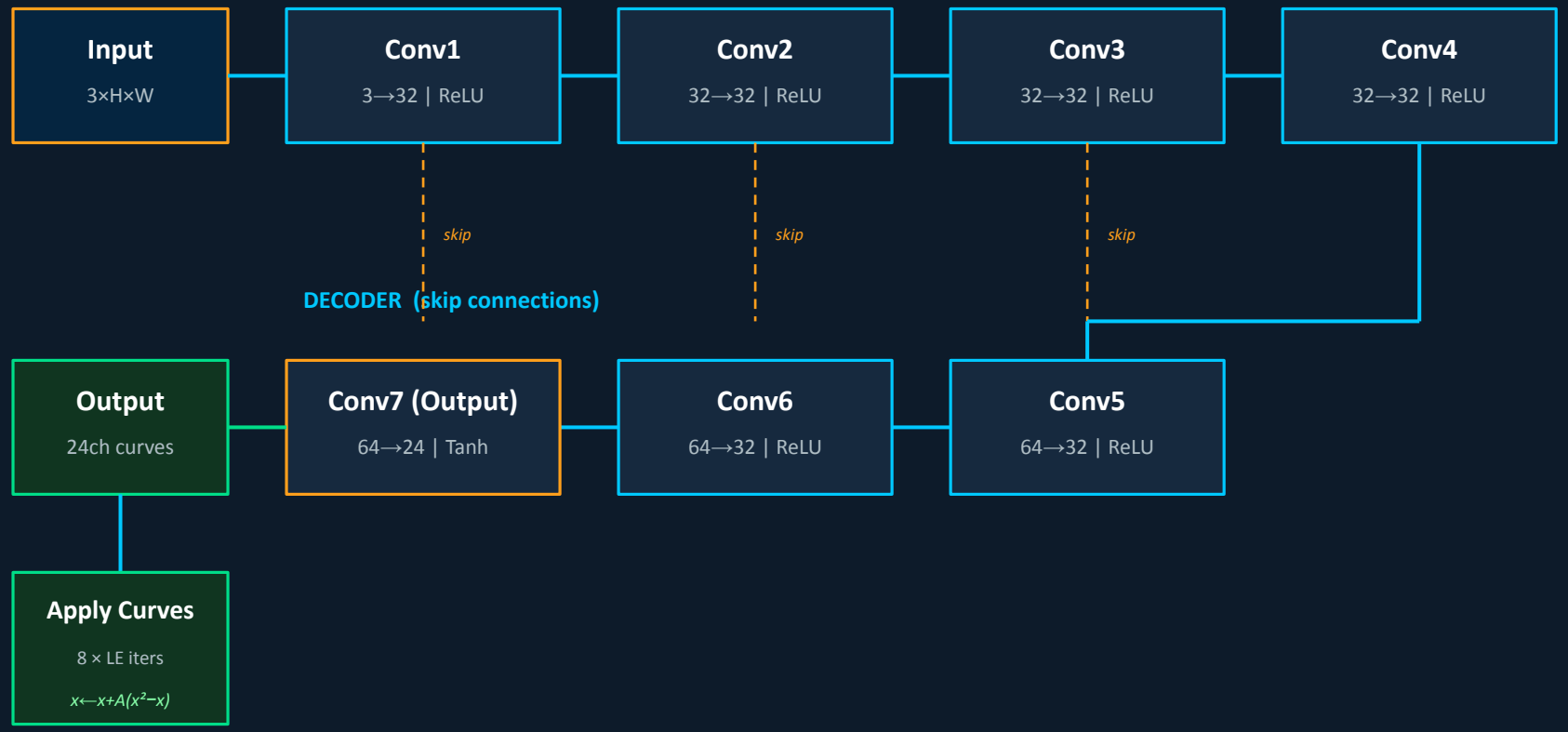
Digital cameras
in the dark
69 images

NPE

Naturalness-Preserved
Enhancement
84 images

Training split: 80% Train • 10% Val • 10% Test | Batch: 16 • Resolution: 256×256

DCE-Net Architecture



Activation

ReLU (encoder)
Tanh (output)

Channels

32 per layer

Kernel

3×3, stride 1, pad 1

Parameters

~0.08 M (lightweight)

Inference

Full res, real-time

Loss Functions (Unsupervised)

Illumination Smoothness

W = 200

Penalises abrupt pixel-level changes in the predicted curve maps, encouraging spatially smooth and stable tonal adjustments across the image.

↳ *Total Variation on 24-channel curve maps*

Spatial Consistency

W = 1

Ensures local neighbourhood relationships (left/right/up/down) in the enhanced image closely mirror those in the original, preserving structure.

↳ *4-directional conv on avg-pooled patches (4×4)*

Color Constancy

W = 5

Minimises differences between per-channel spatial means, preventing colour casts and keeping white-balance neutral after enhancement.

↳ *Gray-World assumption on R, G, B means*

Masked Exposure

W = 10

Drives patch-mean luminance toward 0.6 (well-exposed), but masks out patches already bright in the original to protect streetlights & highlights.

↳ *16×16 avg-pool + brightness mask (threshold 0.6)*

$$\mathcal{L}_total = 200 \cdot \mathcal{L}_illum + 1 \cdot \mathcal{L}_spatial + 5 \cdot \mathcal{L}_color + 10 \cdot \mathcal{L}_exposure$$

Training Setup

Hyperparameters

Optimizer	**Adam**
Learning Rate	**1e-4**
Batch Size	**16**
Epochs	**100**
Image Size	**256 × 256**
Curve Iters	**8**
MEAN_VAL	**0.6**
Num Workers	**2 (pin_memory=True)**

Training Pipeline

① Auto-scan Kaggle input for images (PNG / JPG / JPEG)

② 80/10/10 train-val-test split (seed=42)

③ Resize to 256×256 ⟶ ToTensor ⟶ DataLoader

④ Forward: DCE-Net predicts 24-ch curve maps

⑤ apply_curves() iterates 8× tonal adjustment

⑥ Compute 4 losses ⟶ back-prop ⟶ Adam step

Inference Pipeline

① Load best .pth (map to CUDA / CPU)

② Resize frame to 512×512 ⟶ ToTensor

③ model(frame) ⟶ curve_maps

④ TEMPORAL SMOOTH: $\alpha \cdot new + (1-\alpha) \cdot prev$ (α=0.15)

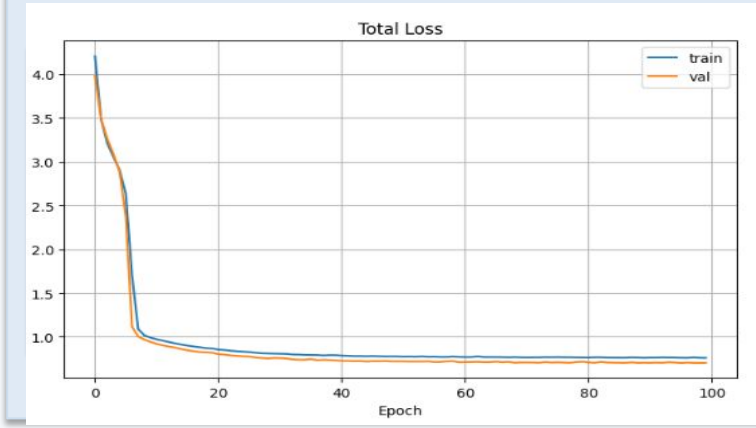
⑤ apply_curves ⟶ clamp(0,1) ⟶ resize back

⑥ Write BGR frame to mp4 (mp4v codec)

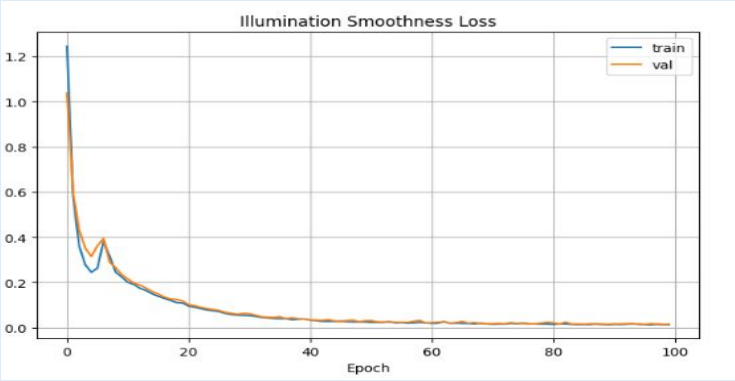
⑦ Zip all enhanced clips for download

Training Curves & Results

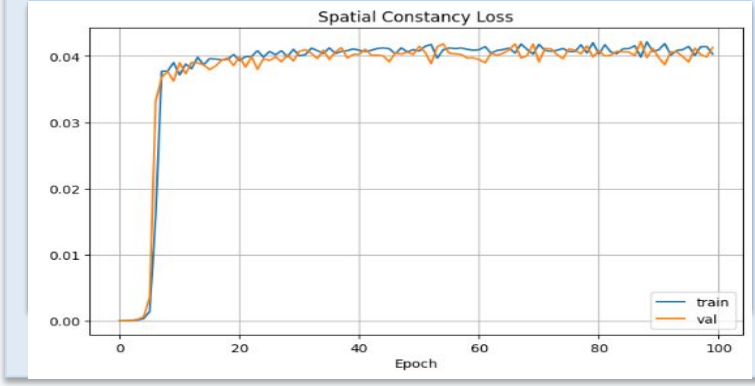
Total Loss



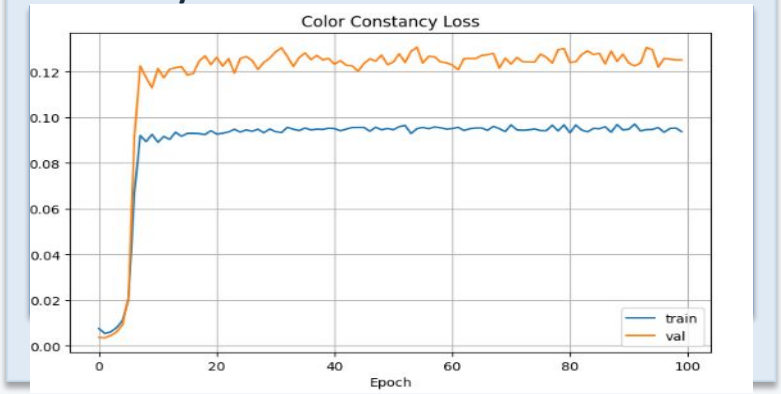
Illumination Smoothness



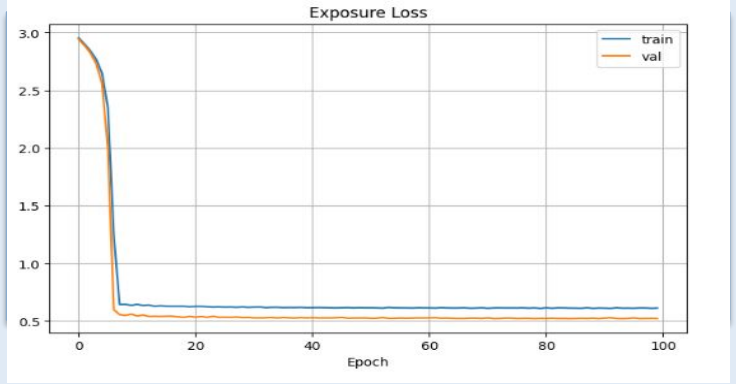
Spatial Consistency



Color Constancy



Exposure Loss



Key Contributions & Conclusion

Zero-Reference Learning

Eliminates the need for paired training data by using four physics-inspired loss terms to self-supervise the network.

Lightweight DCE-Net

~0.08 M parameters — fast enough for real-time video inference on Kaggle T4 GPUs.

Temporal Smoothing

Exponential blending (α=0.15) of curve maps across frames removes flickering in enhanced video output.

Highlight Preservation

Exposure loss uses a brightness mask to avoid over-brightening already-lit regions like streetlights.

Multi-Dataset Training

Fusion of 7 diverse low-light datasets (LOL, LIME, LSRW, MEF, LOL-v2, DICM, NPE) improves generalisation.

End-to-End Pipeline

Full pipeline from raw dataset to enhanced MP4 video, with automated zip packaging for download.

"Illuminate the unseen — without a single ground-truth label."